1 $\sqrt{5^2 + (-12)^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

[Total Marks: 2]

2 (a) Determines column vector $\mathbf{a} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ **[M1]**

10 **[A1]**

(b) Determines column vector $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ **[B1]**

Calculates resultant vector $2\begin{pmatrix} -6 \\ 8 \end{pmatrix} + 3\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -12 \\ 16 \end{pmatrix} + \begin{pmatrix} 9 \\ -12 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ **[M1]**

5 **[A1]**

[Total Marks: 5]

3 (a) $\sqrt{(-8)^2 + 15^2}$ or $\sqrt{64 + 225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p} + 2\mathbf{q} = \begin{pmatrix} -8 \\ 15 \end{pmatrix} + \begin{pmatrix} 4 \\ -8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2 + 7^2}$ or $\sqrt{16 + 49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Total Marks: 5]

4 (a) Sets up equation: $p^2 + p^2 = 18$ **[M1]**

3 **[A1]**

(b) Finds expression for $\mathbf{w}$: $\begin{pmatrix} 4 - t \\ 1 + 2t \end{pmatrix}$ **[M1]**

Forms magnitude squared equation: $(4 - t)^2 + (1 + 2t)^2 = 29$ **[M1]**

Expands brackets correctly: $16 - 8t + t^2 + 1 + 4t + 4t^2 = 29$ **[M1]**

Rearranges to a 3-term quadratic: $5t^2 - 4t - 12 = 0$ **[M1]**

$t = 2$, or $t = -1.2$ (or $-\frac{6}{5}$) **[A1]**

[Total Marks: 7]

5 (a) $\sqrt{x^2 + (x + 2)^2} = 10$ or $x^2 + (x + 2)^2 = 10^2$ **[M1]**

$x^2 + x^2 + 4x + 4 = 100$ **[M1]**

$2x^2 + 4x - 96 = 0$ leading to $x^2 + 2x - 48 = 0$ with no errors seen **[A1]**

(b) $(x + 8)(x - 6) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -8, x = 6$ **[A1]**

(c) $\mathbf{u} = \begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Total Marks: 7]

6 (a) $\vec{AC} = \begin{pmatrix} 7-3 \\ 1-(-2) \end{pmatrix}$ or $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB} = \begin{pmatrix} -8 \\ y+2 \end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2 + (y+2)^2} = 17$ **[M1]**

$64 + (y+2)^2 = 289$ or $(y+2)^2 = 225$ **[M1]**

$y + 2 = 15$ or $y + 2 = -15$ **[M1]**

$y = 13$, $y = -17$ **[A1]**

[Total Marks: 7]

7 (a) Expresses vector $\vec{PQ}$ as $\begin{pmatrix} k-2 \\ 5-(-3) \end{pmatrix} = \begin{pmatrix} k-2 \\ 8 \end{pmatrix}$ **[M1]**

Sets up magnitude equation: $(k-2)^2 + 8^2 = 10^2$ **[M1]**

Simplifies to solve for k: $(k-2)^2 = 36$ **[M1]**

$k = 8$, or $k = -4$ **[A1]**

(b) Finds $\vec{QR}$ using $Q(8, 5)$ and $R(4, 1)$: $\vec{QR} = \begin{pmatrix} 4-8 \\ 1-5 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$ **[M1]**

Calculates magnitude: $\sqrt{(-4)^2 + (-4)^2} = \sqrt{32}$ **[M1]**

5.66 **[A1]**

[Total Marks: 7]

8 (a) $\begin{pmatrix} 9-2 \\ 4-(-3) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2 + 7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2 + 4^2} = 5$ or $k^2 + 4^2 = 5^2$ **[M1]**

$k^2 = 25 - 16$ or $k^2 = 9$ **[M1]**

$k = 3$, $k = -3$ **[A1]**

[Total Marks: 7]

9 **(a)** $\mathbf{u}+\mathbf{v}=\begin{pmatrix}3k+1\\k-1\end{pmatrix}$ **[M1]**

$(3k+1)^2+(k-1)^2=(\sqrt{50})^2$ **[M1]**

$9k^2+6k+1+k^2-2k+1=50$ **[M1]**

$10k^2+4k+2=50 \implies 5k^2+2k-24=0$ **[A1]**

(b) $(5k+a)(k+b)=0$ where $ab=-24$ or $5b+a=2$, or correct substitution into quadratic formula **[M1]**

$(5k+12)(k-2)=0$ **[M1]**

$k=2$ **[A1]**

$k=-2.4$ or $-\dfrac{12}{5}$ **[A1]**

(c) Substituting $k=2$ and $k=-2.4$ into $\begin{pmatrix}2k\\k-3\end{pmatrix}$ **[M1]**

$\begin{pmatrix}4\\-1\end{pmatrix}$ and $\begin{pmatrix}-4.8\\-5.4\end{pmatrix}$ **[A1]**

[Total Marks: 10]